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COLLEGE OF ENGINEERING
KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

AE 464

CROP STORAGE AND STORAGE STRUCTURES

PROF. A. BART-PLANGE

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Introductory Concepts

1.1 Why storage of Agricultural Produce?

- Crop storage is a subject of particular relevance at a time when so many countries are struggling to overcome the effects of natural disasters.
- The only hope for survival, for people and animals, is what's in store.

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1.2 The importance of storage

- They help to ensure household and community food security until the next harvest and commodities for sale can be held back so that farmers can avoid being forced to sell at low prices in the glut that often follows a harvest.
- While considerable losses can occur in the field, both before and during harvest, the greatest losses usually occur during storage.

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Durable Crop Storage





The security of your grain is as good as your storage structure and the quality of its management

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1.3 The basic objective of good storage

- Creation of environmental conditions which protect the product and maintain its quality and its quantity, therefore reducing product loss and financial loss.

Storage is thus useful for the following reasons:

- For future use
- To prevent waste
- For higher prices
- Taste and quality enhancement
- For planting materials for the next season
- Lack of ready market
- Means of savings
- Cultural practices
- Inaccessible roads
- Research/ germplasm

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1.4 Post-harvest losses in grains

Definition of terms

- **Loss**- denotes a measurable decrease of food (grain) value which may be quantitative, qualitative, nutritive or economic.
- **Damage-** refers to the superficial evidence of deterioration (e.g. holed grain by insects pests, or broken grain as a result of threshing from which loss may result).

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Categories of losses

- **Quantitative loss** – denotes a reduction in weight of a product and can therefore be readily defined, qualified and valued.
- **Qualitative loss –** represents damage or contamination of the grain. It is usually based on subjective judgements and therefore difficult to assess or to quantify. However, it can be described by comparison with locally acceptable quality standards.
- **Nutritional loss-** denotes a reduction in the food value of the crop. This may be a result of quantitative or qualitative loss on the product and is also difficult to quantify.

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Categories of losses

- **Germination loss –** denotes a reduction in germination ability of a seed sample due either to the quantitative or qualitative loss or deterioration. Laboratory methods do exist to precisely determine the degree of viability of a given sample of seed.
- **Economic loss –** denotes a reduction in monetary value of the product as a result of loss either to the quantitative or quality of the products.

Loss in quality

- **Quality loss** may occur during processing and storage but in storage they occur from:

i. Moisture build up
ii. Attack from Insects, rodents, birds, micro organisms
iii. Weather effects
iv. Wind

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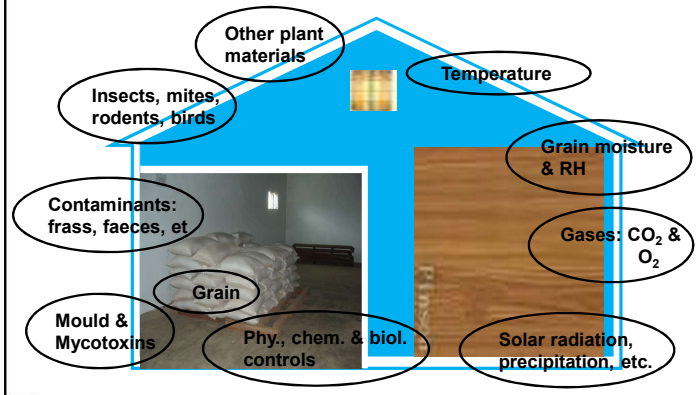
Loss in quantity

- Loss shows up as a loss of weight of the foodstuff due to it being eaten by insects, rodents and birds or to drying of the product.

- These weight losses are not always visually apparent. For example, some insects will eat only the centres of grain kernels so, even though the volume of grain may appear to remain the same, there can be considerable weight loss.

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The Stored Grain Ecosystem



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Factors that affect Grain Spoilage in Storage

- Grain deterioration in a stored grain ecosystem depends on

i. Stored grain moisture
ii. Stored grain environment
❑Temperature and Relative Humidity (RH)
❑CO_2 and O_2
iii. Initial storage mould infestation
iv. Initial insect pest infestation and insect pest type
v. Grain quality (presences of fines/intact kernels)
vi. Sanitation and storage structure
vii. Stored grain management

1.5 Factors for food deterioration in storage

❑**Physical Factors**
- Moisture content of the product (humidity)
- Temperature – grain temperature and ambient temperature

❑**Biological factors**
- Vertebrates – rodents, birds
- Arthropods - insects
- Micro fauna – bacteria, moulds, etc

❑**Technical factors**
- Storage conditions
- State of the product

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Basic Concepts of Stored Grain Management 1

- **Cereal grains and oilseeds are hygroscopic.** They lose moisture when exposed to dry air or gain moisture when exposed to wet air until equilibrium is reached.
- The weather, primarily the air vapour mixture properties determine strategies for drying and storing a particular grain type.

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Basic Concepts of Stored Grain Management 2

- Grain is a biologically active material
- Special care needed to prevent quality losses
- Stored grain is a man-made ecosystem which attracts pests and rodents which can reduce quality.
- Exclude and manage pests
- Grain quality WILL NOT IMPROVE during storage
- Important 1st step: understand the history and initial quality of stored grain

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Important parameters of air for storage

- **Dry Bulb Temperature:** is the air temperature measured by using a thermometer or temperature sensor.
- **Relative Humidity (RH)**: for a given temperature, the RH is the ratio of the vapour pressure of the water vapour contained in an air-vapour mixture to the vapour pressure of an air-vapour mixture that is completely saturated (100% RH).

- **Equilibrium Moisture Content (EMC)** for a grain is the moisture content at which the grain will attain (equilibrium) after being exposed to a given ambient condition.

- The grain equilibrates to the ambient relative humidity at the given ambient temperature.

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Good storage thus involves controlling the following factors within the storage environment:

i. Temperature
ii. Moisture
iii. Pests
iv. Hygiene

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Temperature

- The temperature within a store is affected by the:

➢ sun,
➢ cooling effect of radiation from the store,
➢ outside air temperatures,
➢ heat generated by the respiration of both the food in store and
➢ any insects present.

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Temperature

- With a few exceptions, micro-organisms thrive in environments with temperatures between 10 and 60°C.
- Insects have a narrower range which generally lies between 16 and 45°C.
- As stores in most parts of the tropics and sub-tropics have temperatures between 25 and 35°C, the effects of both micro-organisms and insects are obviously very important.
- Direct temperature control of small stores is not usually a technical or economic possibility so other measures, particularly reducing the moisture content of the stored produce, are necessary.

Temperature

- Sunning: It involves the product, such as grain, being laid out in a thin layer in the hot sunlight. The insects tend to leave the grain if the temperature reaches 40-45°C. It should be noted that sunning does not always destroy eggs or larvae.
- Incorrect store temperatures can result in biological and chemical damage. Examples include the loss of germination ability and the accumulation of sugars in commodities such as potato and sweet potato which need relatively low storage temperatures.

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Temperature

- The speed of chemical change in a food depends upon the temperature and the food's moisture content. A 10°C rise in temperature causes an approximately two-fold increase in the fate of reaction.
- Thus, cold storage will retard such changes as fat oxidation and vitamin loss. Many dried foods benefit from even a small reduction in their storage temperature, and cool and dry conditions can greatly reduce the rate of development of brown discoloration and off-flavours.

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Temperature

- Incorrect temperature control can result in physical damage.
- At high temperatures, fats in products may melt, while at low temperatures, sugar in sweet foods may crystallize.
- Chilling injury may occur. Considerable tissue disruption may occur when foods are frozen.
- Freezing also destroys the stability of emulsions, and foods such as milk and cheese may show a marked separation of fat upon thawing.

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Moisture

- All micro-organisms, including moulds, require moisture to survive and multiply.
- All materials that have been dried will try to come back into equilibrium with the climate around them and in tropical countries this usually means absorbing moisture.
- The moisture level below which micro-organisms cannot grow is referred to as the safe moisture content.

Table 1. Safe moisture content of stored products for long storage

Product	Safe moisture content%
Maize	13
Millet	13
Sorghum	13
Rice	13
Cowpea	15
Groundnut	7
Cocoa	7
Palm kernel	5
Coffee	13

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Moisture

- The siting and the ventilation of the store are important. If the walls of a store are cooled below their dew point by low night temperatures, condensation can occur and increase the moisture in the layers of produce near the edge of the store.
- It is important to remember that most stored food products are 'alive' and respiring, giving off moisture as well as heat.

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1.6 Presence Of Moisture In Stored Grain

Starch + Oxygen→ CO_2+H_2O+Heat

- Accumulation of heat can lead to mould formation in the silo.
- The quality of the grain is reduced
- The viability of the grain is reduced
- Respiration rate increases

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1.7. Physical factors in storage

- Temperature of the air
- Relative humidity of the air
- Moisture content of the stored product

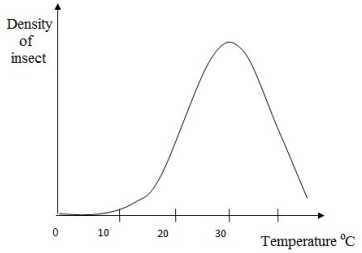


Fig. 1 Effects of temperature on insect multiplication in grain storage

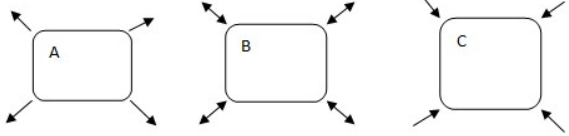


Fig.2. Effects of relative humidity (RH) on moisture absorption

A - Low relative humidity; air absorbs moisture from the grain hence leading to drying of the grain

B - Equilibrium is established

C - High relative humidity; the grain absorbs water from the air. Hence there is moistening of the grain

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PEST

Keeping pests out of the store

Mammals

- The most serious pests for store owners are rodents.
- All holes and entrances to the store should be fitted with fine wire mesh.
- The building should, if possible, be on poles with rat baffles around them as shown in the diagram.
- If the building is situated under a tree, rats can readily gain access on to the roof.

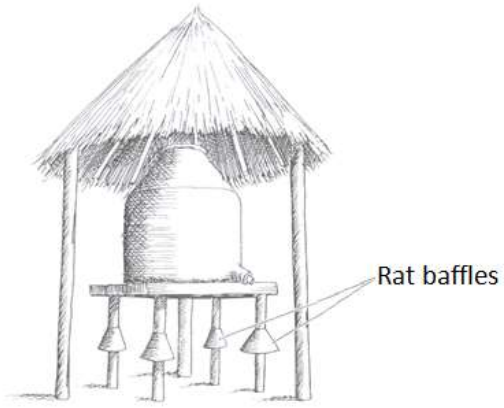
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Keeping pests out of the store

Mammals

- The ground around the store should be cleared and all sources of water denied to rodents.
- They can easily gnaw through wooden doors so tin plate should be fixed to the bottom 30cm of the door.
- Domestic animals such as goats and poultry can pose a problem and should be kept at a distance from the store by building simple protective fences.

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Food store showing rat baffles

Insects

- The main purpose of insect control is to prevent insects entering the storage containers.
- The containers should be closed tightly; baskets may be smeared with mud to seal them.
- Access to the store by insects such as ants and termites can be prevented by building the store on poles which are treated with coal tar, green camphor oil, or covered with fat or waste oil.
- Surrounding the supporting poles with a layer of sieved wood ashes, and sinking the poles into ground soaked with waste oil, will also deter insects.

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Insects

- The smoke and heat of a fire will kill or drive insects away from stored produce. For this reason, many traditional house-hold stores are built in the roof of the house or over the cooking fire.
- Mixing the produce with materials such as wood ash, burnt cow dung, fine sand, lime and certain clays deters insects, mainly by scratching their bodies, which causes them to lose water and die.
- Certain vegetable oils such as palm oil and groundnut oil can be applied to pulses to give protection against bruchids (beetles).

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Insects

- Roots, leaves, fruits and flowers of some local plants are also known to act as an insect repellent. Notable example is the neem leaves.
- Husks, pods and outer coverings offer some protection against insect attack provided they are undamaged. Traditional maize varieties often have a husk which covers the whole cob giving better protection than improved varieties which do not always have such a full husk.

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Insects

- The storeroom should always be kept as clean as possible.
- All material entering the store should be checked to see that it is not infested, and put into containers only after all the old product, dust, straw, etc. have been removed.

Control of pests in stores

- Rodent poisons should be placed in containers and never allowed to come into contact with the food. The poisons should be changed regularly to avoid the pests developing resistance.
- It should be remembered that rodents, after eating some types of poison, will often hide away and die. Other poisons make the animals very thirsty and so they leave the store in search of water.

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Control of pests in stores

- In view of the dangers associated with rodent poisons, if their use is not properly controlled, consideration should be given to simpler traditional measures such as the use of cats, though their effectiveness is a matter of some debate.
- In larger stores, electric 'insect-catchers' are often used to control flying insects such as moths.
- The use of chemicals or traditional methods, while important, is no replacement for good store management and regular inspection.

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The Concept of Grain Storability

- Rate of grain dry matter loss (quality and grade loss) due to growth of grain storage fungi/insects.
- Storage time to 0.5% dry matter loss is called the allowable storage time (AST).
- Grain storability is affected by prior post-harvest handling, kernel damage, storage conditions (moisture and temperature) and other factors.

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Storage Life of Grain

°C	Corn temperature ° F	Maximum storage time (months) for corn and soybean* Moisture Content Corn *(top %)*, Soybean *(bottom%)* 13%, 11%	14%, 12%	15%, 13%	16%, 14%	17%, 15%	18%, 16%	24% N/A
5	40	150	61	29.0	15.0	9.4	6.1	1.3
10	50	84	34	16.0	8.9	5.3	3.4	0.5
16	60	47	19	9.2	5.0	3.0	1.9	0.3
21	70	26	11	5.2	2.8	1.7	1.1	0.2
27	80	15	6	2.9	1.6	0.9	0.9	0.06

*Based on 0.5% maximum dry matter loss—calculated on the basis of USDA research at Iowa State University. Corresponds to one grade number loss; 2-3% points in damaged seeds. Soybean approximated at 2% lower moisture than corn.

Losses caused by mould growing on grain

- Decreased germination
- Discoloration of the germ
- Heating or mustiness
- Decrease in nutrient value
- Mycotoxins (***such as aflatoxin)*** *can be* produced by some moulds
- Loss of weight



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Grain Losses Caused by Insects

Insect damage in a stored grain ecosystem depends on

- Initial insect pest infestation
- Grain nutritional value
- Grain moisture
- Moulds
- Presence of predatory insect
- Stored grain environment
  - ❑Temperature and RH
  - ❑CO_2 and O_2

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2. DURABLE CROP STORAGE

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Grains



- *Most often, the value of the grain (in dollars-per-tonne) is usually greater than the cost of the structure in which it is stored. Minor expenditure in improving the quality of the storage can thus be quickly recovered if commodity losses are commensurably reduced.*

- Compared to most other foodstuffs, such as meats and vegetables, grains are relatively easy to store. **If grain is kept insect-free and below its safe moisture content**, it will keep for many years with **minimal loss of quality** or nutritional value.

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The choice of storage design is wide, but the **essential requirements needed to store grain safely are** the same.

- These are:

- **Sealing.** The storage structure must keep the grain **free from water ingress, insects, rodents and birds**.
- **Cooling.** If grain is to be stored at moisture content above 'safe' levels, provision should be made for **cooling the grain**.
- **Easy disinfestation.** The storage facility should also permit easy and economical **disinfestation of grain** in the event of insect infestation.

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Durable Crop Storage

- Storage depends on:

1. Type of crop
2. Nature of the crop at harvest
3. Total volume of crop
4. Storage requirement for the crop
5. Form in which the crop is to be stored
6. Cost of storage

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2.1 Properties of stored products

- ❖Bulk density – It varies with the moisture content
- ❖Angle of repose – It increases with the moisture content
- ❖Static coefficient of friction
- ❖Particle density
- ❖Porosity

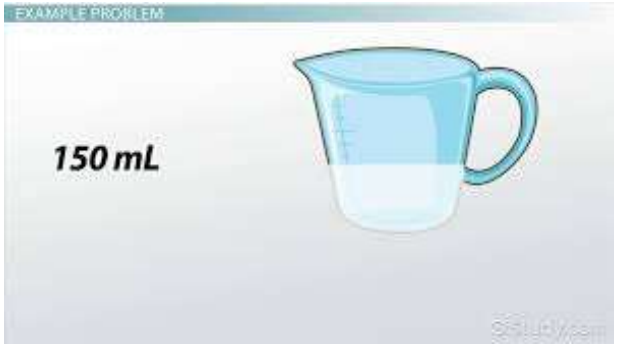
2.2 Types of Durable Storage

- Bulk storage
- Sack/bag storage

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Bulk density



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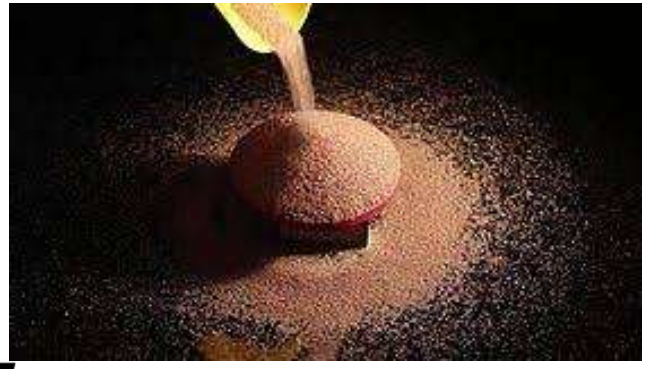
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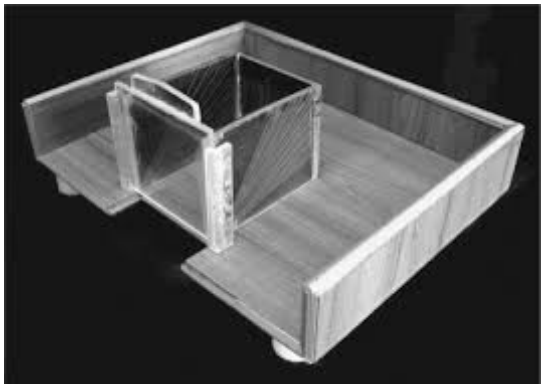
Angle of repose



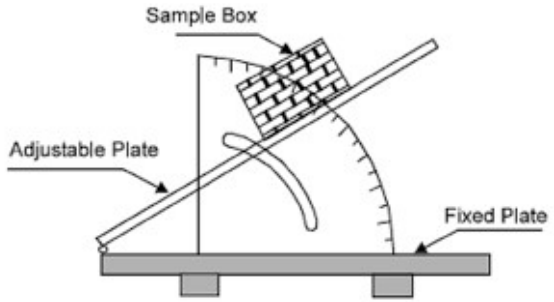
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Angle of repose



Static coefficient of friction



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3.1 Bulk Storage

- Bulk storage can be grouped into traditional and conventional/ modern

Bulk storage techniques:

- ❖Non-hermetically sealed storage
- ❖Hermetically sealed storage

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Hermetic Storage

What is hermetic storage?: a technique for storing grain in a sealed air-tight container causing the living organisms (grains and insects) to deplete the oxygen and increase carbon dioxide of the enclosed microenvironment by their respiration to levels that will cause insects and mold not to survive.

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Mechanism of Oxygen Depletion in Hermetic Micro-environment



Hermetic Storage Systems



GrainPro Cocoon™



Purdue Triple Inner-Bag (PICS)



SuperGrainBag
GrainPro, Inc.

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REQUIREMENTS OF AN IDEAL GRAIN STORAGE STRUCTURE

- The object of an ideal **grain storage structure** is to control and reduce the **storage** loses from rodents, insects and micro-organisms, birds, moisture and heat to a minimum
- A good storage structure is the one, which can provide protection against all possible causes of damage. A food storage structure, for storing food grains on a large scale, should have the following essential features:
-

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- It should be easy to clean.
- 2. It should provide protection from rodents, birds and other animals.
- 3. It should be waterproof and moisture proof.
- 4. It should protect the food grains against variations of temperature and humidity.
- 5. It should have provision for periodical inspection.
- 6. It should have provision for application of pesticides through spraying or fumigation.

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- It should be located far away from possible sources of infection such as kilns, flour mills, and bone crushing mills, garbage rumps, tanneries, slaughter houses and chemical industries.
- 8. It should be located at a convenient place from where it is easy to receive issue and transport the food gains. This explains why most of the storage structures are located near railway stations or on highways.

Therefore in designing and constructing storage structure following points shall be borne in mind:

- All holes, pipes and ducts and other openings shall be guarded by suitable means, such as gratings, etc., in order to prevent the entry of rats and other vermin.
- The structure shall have smooth, crack free internal surfaces and shall have no unnecessary cavities and projections to prevent the lodgement from insects and vermin. Periodical fumigation and other treatments should be done to eliminate infestation of grains by insects, fungus etc. The structure shall be designed so as to facilitate its sealing for fumigation or have facility to seal a portion where fumigation has to be carried out, or it may be made completely airtight if required.

- Structures should have good ventilation arrangement to prevent moisture accumulation in pockets.
- The structure shall be designed to make it possible to control moisture. Moisture may be controlled by adopting methods of construction using non-hygroscopic material, by sound wall, roof and floor construction, by the use of vapour barriers, and by the use of aeration.
- The structure shall be so oriented that it will receive the minimum solar radiation. Reflective external surfaces, insulating materials, sun shades, a minimum of glass surfaces, controlled ventilation and aeration, to reduce the internal temperature may be used.

3.2 Traditional bulk storage methods/structures

- *3.2.1 Earthenware pots and gourds*
- In the tropics, earthenware pots and gourds (the hard dried outside skins of certain fruits and vegetables) are widely used for storing small quantities of foodstuffs.
- They need to be kept sheltered: above the kitchen where there are few insects is a good place.
- Such containers should be raised off the ground or put on plastic sheets to stop them absorbing moisture from below.

3.2.1 Earthenware pots and gourds

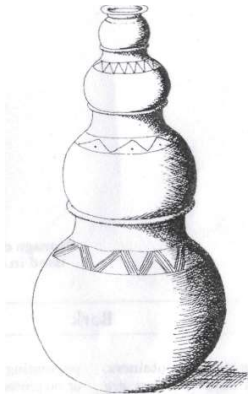
- The storage characteristics of earthen- ware pots and gourds can be improved by treating them with
  - ✓varnish,
  - ✓paint, or oil, and
  - ✓by using sealed lids.
- The contents should be regularly inspected to check for mould growth and infestation.
- If moisture has been absorbed then the food stuff should be removed and redried.

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3.2.1 Earthenware pots and gourds

- Suitable for small quantities of:
  - ➢cereals,
  - ➢beans,
  - ➢groundnuts,
  - ➢dried fruit and vegetables and
  - ➢seed material
- *Storage time:* up to one year
- *Capacity:* 5—30litres
- *Cost:* very low

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Earthenware pots

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3.2.2 Baskets

- Baskets are widely used throughout the tropics for food storage.
- For certain products, good ventilation is required, so the baskets should not be placed too close together and if possible they should be raised off the ground.
- Woven baskets do not give much protection against insects.
- Baskets can be covered in plaster, mud or clay on both the inside and outside.

3.2.2 Baskets

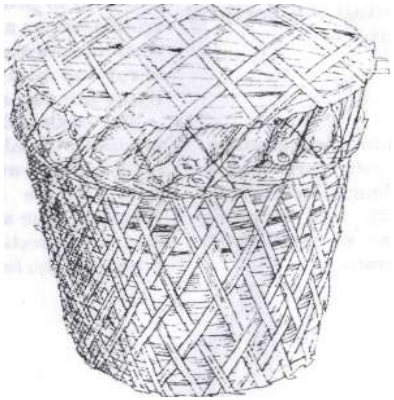
- This will improve protection against insects but reduce ventilation.
- Covered baskets are suitable for products that do not require ventilation.
- There is always a danger, however, of this covering cracking and then harbouring insects.
- Some improvement can be achieved by using a plastic sheet lining inside the basket.
- Food stored in baskets should be well protected from rain, preferably in a dry area of the house or building.

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3.2.2 Baskets

- Suitable for:
  - ➢cereals,
  - ➢pulses,
  - ➢oilseeds,
  - ➢potatoes, etc.
- *Storage time:* up to nine months
- *Capacity:* variable
- *Materials:* reeds, grasses, palm leaves, bamboo, etc.
- *Cost:* low, although considerable labour may be involved

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Baskets are widely used for food storage

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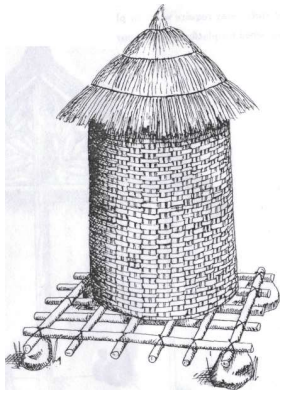
3.2.3 Basket Silos

- A basket silo *is* cylindrical and *is* made of woven sheets of plant material such as elephant grass or reeds.
- Small silos may have a capacity of 100kg of grain with larger ones able to hold over half a tonne.
- Basket silos may be plastered with mud for extra protection.
- If they are not plastered on the inside, they are sometimes lined with straw or hessian.
- Such silos are reasonably durable and may be re-used from one season to the next.

3.2.3 Basket Silos

- *Suitable for:* cereals and pulses
- *Storage time:* up to one year
- *Capacity:* up to a tonne
- *Materials:* elephant grass, reeds, sorghum stalks
- *Cost:* local material, time spent on construction

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Basket silo on elevated platform for improved aeration

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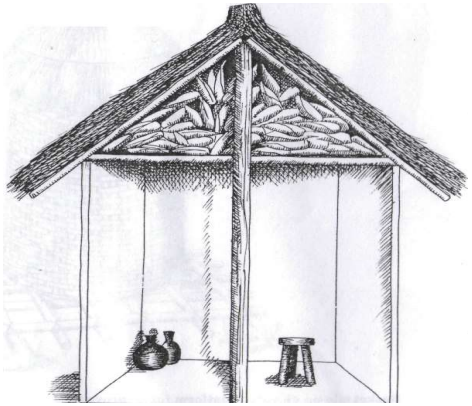
3.2.4 Roof storage

- Many foods, particularly cereals, are stored indoors in the roof of the house above the kitchen hearth.
- Sometimes special platforms are constructed.
- Heat and smoke from the fire helps to keep the product dry and to deter insects.
- One disadvantage is that it is difficult to protect the foodstuffs from rodents.
- Nevertheless the technique has considerable advantages in terms of low cost, control of insects and maintaining the cereals in a good dry condition.

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3.2.4 Roof storage

- *Suitable for:* cereals
- *Storage time:* up to one year
- *Capacity:* variable
- *Materials:* may require wood for platform
- *Materials:* may require wood for platform
- *Cost:* wood for platform and labour



Roof storage

3.2.5 Maize cribs

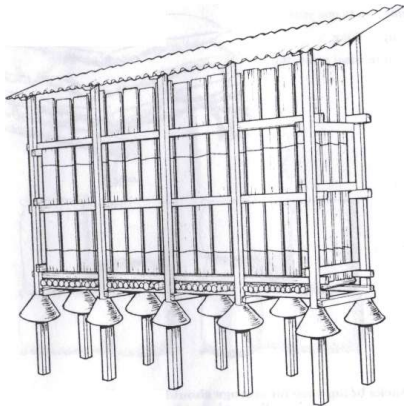
- Traditional maize cribs are suitable for the storage of maize cobs in semi-humid and dry tropics.
- The shape of the crib allows drying to proceed during storage using natural ventilation, provided that relative humidities are below 70 per cent.
- Cribs are preferably sited with their long side to the prevailing wind direction.
- Because of the narrow shape, usually 60-90cm in width, the drying process is better than in some traditional round cribs.

3.2.5 Maize cribs

- Because of the narrow shape, usually 60-90cm in width, the drying process is better than in some traditional round cribs.
- Little protection is provided against insects.
- Traditional maize varieties with husks that cover the whole cob are protected reasonably well for three to six months.
- Removal of husks speeds up the grain drying and after reaching a safe moisture level, the maize is often shelled and stored in a less bulky way.

3.2.5 Maize cribs

- *Suitable for:* maize
- *Storage time:* up to six months
- *Capacity:* variable
- *Materials:* variable
- *Cost:* labour and materials



Maize crib

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3.2.6 Underground pits

- Storage in underground pits is a widely practised method in many tropical areas.
- Pits are cheap to construct, provided here is a sufficient depth of workable soil.
- They are of particular use in dry areas where wood for normal store construction may be in short supply.
- Traditional pits provide some protection from both insect and rodent attack, although termites can pose a problem.

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3.2.6 Underground pits

- Mould damage often occurs in the vicinity of the lit sides and on the surface, due to migration of moisture from the soil and atmosphere.
- Good pit storage practice therefore depends on restricting the supply of air and moisture moving into the pit from the surrounding soil or from the atmosphere.

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3.2.6 Underground pits

- Traditional pit lining is carried out with plant materials (such as grass, straw and chaff or maize stalks), clay, and cow dung or termite-mound soil.
- If only plant material is used, it probably does little more than keep the grain from direct contact with the soil around.
- The use of clay reduces water entry but does not prevent it totally.

3.2.6 Underground pits

- After filling, the pits are closed and sealed with plant material and soil and they are usually plastered over.
- *Suitable for:* cereals, pulses and root crops
- *Storage time:* up to one year
- *Capacity:* variable
- *Materials:* grass, straw, chaff and clay
- C*ost:* labour

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3.2.7 Clamp storage

- Root crops and tubers are often cured and stored in field clamps.
- The typical design consists of a circular bed of straw or dried mass approximately 1.5m in diameter and 15cm thick, placed on well-drained round.
- The freshly harvested tubers are heaped in a conical pile weighing 300-30kg on this straw bed and the whole pile is then covered with a similar pile of straw.
- The entire clamp is then covered with soil to a thickness of some 15cm.
- Soil then removed from round the circumfrence of the clamp to give a drainage ditch.
- During cool moist periods, this basic clamp design is satisfactory.

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3.2.7 Clamp storage

- In order to minimize the effects of direct sunshine or heavy rains, clamps may be protected by a thatched roof or positioned under trees.
- In hot dry areas it is necessary to ensure that the internal clamp temperature does not exceed 40°C since tubers will begin to deteriorate rapidly above this temperature.
- Clamps are modified to suit such conditions by applying a thicker soil cover and providing central and bottom ventilation holes to encourage air flow and the outflow of heat.

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3.2.7 Clamp storage

- Such ventilators are constructed from locally available materials such as straw, bamboo, drainpipes or timber.
- Frequent light rainfall can be advantageous since moistening of the soil lowers internal temperatures.
- Gentle wetting of the soil during hot dry periods should be considered.
- If more than 500kg of tubers are to be stored at any one time, it is advisable to use several clamps or one single elongated structure, since internal temperatures are more difficult to control in very large clamps.
- In addition several smaller clamps reduce the possibility of loss of the whole stock by rotting.

3.2.7 Clamp storage

- Before making any specific recommendations for clamp storage, simple trials using locally available materials should be undertaken in order to determine the best design and location of clamps.
- Clamp storage of cassava by inter-layering with cassava leaves, and replacing the straw cover, first with cassava leaves and then with coconut fronds, appears to give better results.
- *Suitable for:* tubers
- *Storage time:* up to six months
- *Capacity:* up to 500kg *Materials:* grass, straw *Cost:* labour

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3.2.8 Earth silos

- Earth (i.e. mud and straw) storage structures are often used in dry tropical areas. They are less suitable for wet regions because water may readily enter via the walls. Earth silos may be made more waterproof by:
- using a mixture of 90 per cent loam or clay and 10 per cent cement for the construction of the walls;
- painting or coating the outer walls with coal tar, asphalt, oils or water-repellent paints;
- applying a layer of waterproof mortar to the walls.

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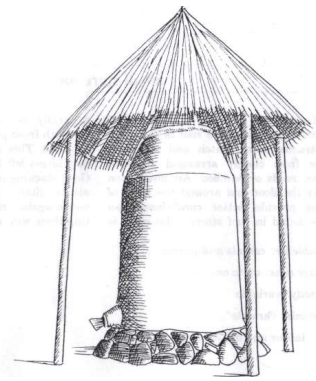
3.2.8 Earth silos

- Whitewashing will help to keep the silo cooler and it will fill in tiny cracks and openings.
- The product to be stored should be fully dried before placing it in the silo as further drying is difficult if not impossible.
- Earth silos give better protection against insects than, for example, maize cribs.
- The silos should be protected from rain, which can cause severe damage.
- They do have the disadvantage of some moisture uptake and a comparatively short life.

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3.2.8 Earth silos

- *Suitable for:* pulses and cereals
- *Storage time:* up to one year *Capacity:* variable
- *Materials:* earth, straw *Cost:* labour



Earth silo

3.3 IMPROVED TRADITIONAL STORAGE STRUCTURES

3.3.1 Plastic bags

- These are widely used for storage in the humid and dry tropics.
- The product has to be dried well because, during storage, further drying is impossible as there is no air circulation.
- When plastic bags are closed well, airtight storage results, with all its advantages and disadvantages.

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3.3.1 Plastic bags

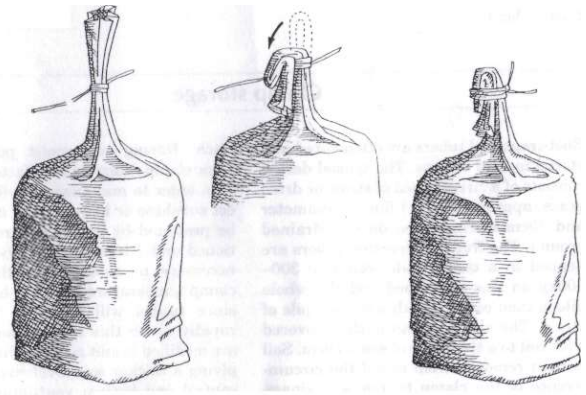
- Plastic bags do not offer much protection against rodents, and they can be pierced by sharp seeds during transport and penetrated by insects.
- This can be reduced by putting a bag of tightly woven cotton inside the plastic bag.
- Plastic becomes weak or brittle after continued exposure to the sun and therefore no plastic package will last indefinitely.

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3.3.1 Plastic bags

- An advantage of transparent plastic is that the product remains visible which makes control more simple.
- Although the product may look good from the outside, however, it may have become musty within. Fertilizer bags cannot be used unless they have been very thoroughly cleaned.
- *Suitable for:* sowing seed, cereals, pulses, groundnuts, copra
- *Storage time:* six to nine months
- *Capacity:* up to 60kg *Cost:* fairly high

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Closing a plastic bag to make it air tight

3.3.2 45-gallon metal drums

- Small drums and water tanks are often available and can be used for storing crops, provided they have been well cleaned.
- When being used for storage, they should not be placed in full sunlight, but protected, preferably under a good roof, and insulated with a layer of straw to prevent large temperature changes.
- A tightly closed drum prevents the entry of insects.
- Grain should be well dried before filling.

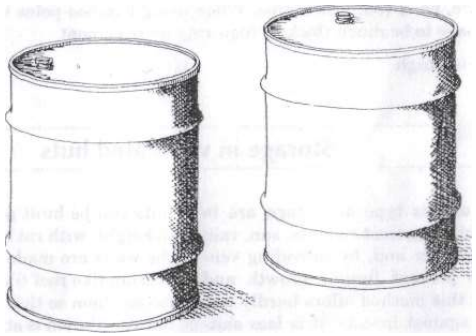
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3.3.2 45-gallon metal drums

- *Suitable for:* cereals, pulses and seeds
- *Storage time:* up to one year
- *Capacity:* 50-200 litres
- *Materials:* oil drums and water tanks
- *Cost:* low, depending on availability

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3.3.2 45-gallon metal drums



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3.3.3 Metal silos

- A whole range of small metal silo designs exists, with silo capacities up to about 5 tonnes.
- Some skill in welding is required to make the structure airtight.
- Silos can be made with overlapping sheets, bolted or riveted together.
- The silo has two openings, one for filling at the top and one for emptying at the bottom.

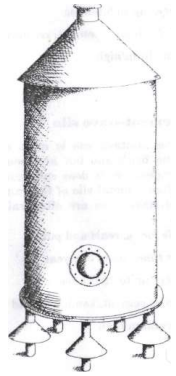
3.3.3 Metal silos

- As in the case of metal drums, metal silos should not be placed in full sunlight, but sheltered to prevent dramatic temperature changes.
- Some small silo designs incorporate a ventilation system operated by natural airflow.
- A rotating fan-like structure is placed on top of the silo and, when the flaps are open, the grain is ventilated by fresh cool air. Metal silos tend to be expensive.

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3.3.3 Metal silos

- *Suitable for:* cereals and pulses
- *Storage time:* approximately one year
- *Capacity:* up to 5 tonnes
- *Materials:* sheet metal
- *Cost:* medium/high,



Metal silo of 3-tonne capacity

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3.3.4 Brick and ferro-cement silos

- Silos of this type are suitable for use in dry and wet tropical areas, but they should be protected from rain by a roof, and the base should be made from reinforced concrete or fired bricks.
- These silos are comparatively airtight and watertight, particularly when painted with rubber-based paints, coal tar or bitumen.
- A whole range of sizes and styles are possible. Three common types are brick, cement-stave and Thai ferro-cement silos

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3.3.4.1 The brick silo

- As the name implies, this silo is made from bricks, placed on a reinforced foundation and covered by a concrete plate with a manhole for filling and emptying. Brick silos are relatively easy to build and can store several tonnes of grain.
- *Suitable for:* cereals and pulses
- *Storage time:* up to one year
- *Capacity:* up to 5 tonnes
- *Materials:* bricks, cement, reinforcing rod, wood for moulds, sheet metal
- *Cost:* medium/high

3.3.4.2 The Thai ferro-cement silo

- Ferro-cement is made of wire mesh (for example chicken-wire), sand and cement and it is strong and durable.
- Ferro-cement silos can be made in almost any shape.
- The Thai ferro-cement silo has a conical shape and is waterproof and airtight.
- The base is saucer-shaped and is made of two layers of reinforced concrete with a layer of bitumen, asphalt paper, plastic or metal foil in between.

- The frame of the walls is made from water pipes or bamboo poles and reinforcing rods, which support an inside and outside layer of wire mesh.
- The mesh structure is filled and smeared with mortar of a paste-like consistency: 1 part of standard cement, 1.75 parts sand with the optional addition of a plasticizer to improve workability.

3.3.4.2 The Thai ferro-cement silo

- *Suitable for:* cereals and pulses
- *Storage time:* nine to 12 months
- *Capacity: 4-6* tonnes, depending on the size
- *Materials:* cement, sand, aggregate, mortar plasticizer, sealant for base, paint, chicken-wire, no. 2 rod, water pipe. When using bamboo poles instead of water pipes, the walls have to be much thicker, requiring more cement.
- *Cost:* medium/high

3.3.5 Storage in ventilated huts

- The aims of this type of storage are to offer protection against rodents, sun, rain and groundwater and, by providing ventilation, to prevent fungus growth and rotting.
- As this method offers hardly any protection against insects, it is less suitable for long-term storage of cereals and pulses.
- This storage method is very suitable for the storage of root crops.

3.3.5 Storage in ventilated huts

- For construction, locally available materials should be used: bamboo, planks, woven mats in a wooden frame, etc.
- If the walls are made of planks they should overlap like roof tiles with some space between them so that sunlight cannot enter but ventilation is still possible.
- The roof of wood or thatch should be overhanging for protection against sun and rain.
- Inside the hut, the products should be piled in boxes or on shelves or racks along the wall, in such a way that air can move freely between them.
- Regular inspection of stored products is necessary

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3.3.5 Storage in ventilated huts

- *Suitable for:* cereals, pulses, root crops
- *Storage time:* variable
- *Capacity:* variable
- *Materials:* local building materials
- *Cost:* medium/high

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3.3.6 Improved pit storage

Roofing

- The roofing of a pit can be made of metal sheet, sealed with mud/dung or bitumen, or polythene sheet.
- A temporary shelter over the pit site gives protection from rain but should be removed in the dry season to ensure drying by evaporation, because a shelter does not prevent lateral movement of water into the pit.

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3.3.6 Improved pit storage

Improving pit linings

- Coating the walls with mud/dung/straw mixture.
- The product is put into well-sealed plastic bags.
- Plastic lining.
- Ferro-cement lining.

- *Suitable for:* cereals, pulses and root crops
- *Storage time:* up to one year
- *Capacity:* variable
- *Materials:* metal sheet, mud/dung/straw or plastic or ferro-cement lining
- *Cost:* medium

4. MODERN OR CONVENTIONAL STORAGE STRUCTURES

MODERN OR CONVENTIONAL STORAGE STRUCTURES

- This category may be classified under:
  - ➢Bulk storage structures
  - ➢Bag storage structures

4.1 Bulk storage structures

- ➢Conventional structures:
  - ✓Reinforced concrete silo
  - ✓Corrugated concrete silo
  - ✓Brick wall silos
  - ✓Plywood silos

4.1.1 Reinforced concrete silo

- Durable, corrosion resistant and to a large extent maintenance free structures.
- These silos typically consists of a cylindrical unit made of reinforced concrete(steel).
- This is necessary to resist the lateral pressures of the grain.
- This type is heavy therefore the foundation must be appropriately constructed.
- There should be assess for loading at the top and unloading the base.
- The sizes range from above 2m. Diameters should sometimes be restricted (1.5m – 4m).

4.1.2 Corrugated steel Silos

- They are usually made of curved sheets of steel bolted together. The joints between the sections are sealed to ensure water tightness.
- They are usually covered by a metal roof. These structures are light and easy to assemble. They are however not able to protect the grains enough from the heat or outside heat.
- They range between 2 tonnes to about 50 tonnes. They can also be combined to have a centralised storage system.

Silo facility



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4.1.3 Advantages and disadvantages of conventional bag and bulk storage

Bags	Bulk
Flexibility of storage	Inflexible storage
Partly mechanisable	Mechanisable
Slow handling	Rapid handling
Considerable spillage	Little spillage
Low capital costs	High capital costs
High operating costs	Low operating costs
Easy inspection	Inspection more difficult

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- The first advantage lies in the occupied ground surface. This method requires **less ground space**, which is important if space is not available or if its cost is high like at ports.
- Another advantage lies in how easy it is to keep the **optimum storage conditions for the grain,** by controlling the temperature, insects, mould, birds, which in long term storage facilities could result in an important economic loss.

- The third advantage consists of **lower costs** than that incurred from using warehouses, **which entails the automation of the grain transport equipment.** The loading as well as the unloading can be completely automated and at a lower cost.
- Another advantage is **lower assembling costs,** since this element is designed specifically for this purpose and weighs less than a storage facility.
- Finally, we have the fact that it is **less expensive overall**, and this is probably the main advantage in comparison to grain warehouses.

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4.2 Silo design considerations

- There are two main considerations in the design of silos:
  - ➢Functional consideration
  - ➢Structural considerations

4.2.1 Functional consideration

- Provision of adequate volume
- Proper protection of the product to be stored
- Satisfactory method of loading and discharging of the product

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4.2.2 Structural Considerations

i. Applicable loads
ii. Materials used for construction of the silo
iii. Dead loads of material
iv. Live loads
  - ➢From the stored products
  - ➢Wind loads
  - ➢Thermal loads
  - ➢Impact and pressures

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Design. When designing a silo plant, we should consider the following criteria:

- It needs to be compact and operational.
- Flexible, allowing alternative processes.
- With open spaces and access, in view of future expansion.
- It should allow the automation of processes, such as grain drying.
- It should have low operating costs (staff, electricity, fuel).
- It should also have a low impact on the environment

To size the facilities we must take into account:

- Production area.
- Crop production and total production.
- Harvest timing.
- Harvest humidity.
- Harvester type and working capacity.
- Number of harvesters.
- Marketing strategy.
- Main objective is: the preparation or storage.
- Internal transport (trucks with trailers, trucks, hoppers).

- Specific weight of most frequently processed grains.
- Processing of seeds.
- Main means of load (truck or rail).
- Quality grading.
- Electricity system.
- Weather conditions (design of ventilation).
- Cost / benefit ratio.
- By evaluating all these items, we can define the number and capacity of the equipment and systems that integrate the silo facility.

4.3 Classification of Silos

- **Orientation**- Silos can be classified as horizontal silos or vertical silos
- **Height**- silos can also be classified as shallow silos or deep silos

4.3.1 SHALLOW SILOS

- The lateral wall pressures increase linearly with depth or height of grain in the silo.

The height (h) ≤ D tan (45^0+θ/2)

θ = angle of internal friction of the grain or the angle of repose

D = equivalent diameter

h = height of grain in silo.

4.3.2 DEEP SILOS

- The lateral pressures relate non-linearly to the height of the grain.

The height, h >D tan (45^0+θ/2)

4.4 TERMINOLOGIES

Angle of repose/angle of internal friction (θ)

- It is the angle formed with the horizontal when a granular material is poured onto a plane surface or onto a close space such as a silo.

- Factors affecting the angle of repose
  - ➢Moisture content
  - ➢Shape of the grain(variety of the grain)

4.4 TERMINOLOGIES

4.4.1 Coefficient of friction/Angle of external friction(μ)

- This is the friction between the grain and the wall. It is the ratio of the vertical wall load to the lateral wall load.
- μ= tan ϕ

4.4 TERMINOLOGIES

4.4.2 Equivalent diameter (ED)

$$ED = \frac{4 \times floorArea}{perimeter}$$

- For circular section, $ED = \frac{4 \times \pi r^2}{2\pi r} = 2r$
- For square section, ED = a; a is the dimension of a side of the square
- For a rectangular section, $ED = \frac{2ab}{a+b}$;a and b are the sides of the rectangular section

4.4 TERMINOLOGIES

4.4.3 Equivalent fluid density (EFD)

$$EFD = w\tan^2\left(45 - \frac{\theta}{2}\right)$$

w = bulk density

θ = angle of internal friction

• EFD can be written as $EFD = w\left(\frac{1-\sin\theta}{1+\sin\theta}\right)$

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Example

A variety of rice has bulk density of 580kg/m^3 and an angle of repose of 36^0. What will be the equivalent fluid density of the rice?

Solution

EFD = 580 kg/m^3 $\left(\frac{1-\sin 36^\circ}{1+\sin 36^\circ}\right)$ = 150.58 kg/m^3

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GRAIN	EFD(kg/m^3)
Shelled maize	580
Rough rice	173
Soya bean	258

Grain	Bulk density(kg/m^3)
Maize	719
Rough rice	578
Sorghum	719
Soya bean	719

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• Co efficient of friction: $\mu = \tan \phi$

Grain	Plywood	Mild steel	Galvanised steel	Concrete
Maize	0.38	0.48	0.30	0.55
Rough rice	0.42	0.45	-	0.52
Sorghum	0.3	0.4	0.3	0.38
Soya bean	-	0.37	-	0.27

4.5 LATERAL LOADS ON SILO WALLS

SHALLOW SILOS

- The lateral loads on the shallow silos may be given by the expression

4.5.1 Lateral unit pressure:

- P_L=EFD x h

P_L – Lateral unit pressure (kg/m^2)
EFD – equivalent fluid density (kg/m^3)
h – height of storage

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4.5.2 Total lateral load

- P_{LT} = EFD x h^2/2

DEEP BINS

- $P_L = wD\left[\frac{1-e^{-4k\mu h/D}}{4\mu}\right]$, D is the equivalent diameter

4.5.3 Vertical loads on shallow walls

- $P_v = \mu P_{LT}$ kg/m N/m
- $P_v = \mu w \frac{(1-\sin\theta)h^2}{2(1+\sin\theta)}$

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4.5.4 Total floor load(P_{ft}) kg or N

- $P_{ft} = W_T - P_{vt}$
- W_T is the total weight of the stored product
- P_{Vt} is total vertical load. It is the product of the vertical load and the perimeter of the cross section of the structure.

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EXAMPLE

- Maize has a bulk density of 750kg/m^3, an angle of repose of 27^0 and an angle of external friction of 24^0.

Determine:

i. The equivalent fluid density
ii. The lateral wall pressures ay 3m away from the base assuming that the height of the silo is 5m.
iii. The total vertical wall pressure
iv. The total floor pressure

The silo is of circular cross section and has a diameter of 4m.

4.6 VOLUME REQUIREMENT FOR SILOS

- The volume requirement for a silo is determined by:
  - ❑Quantity of grain to be stored
  - ❑Type of grain to be stored
  - ❑Required cross section of the silo

- The parameters considered include
  - ❑Total weight of the product
  - ❑Angle of repose
  - ❑Cross section of the silo

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Procedure for determining the volume requirement of a silo

1. Compute the mass of the product in kg. W_T
2. Determine the volume of the product. $V_T = W_T/\rho$
3. Consider the type of silo required either a shallow or deep silo.
4. Determine the equivalent diameter of the silo by relation between the volume of the products and the cross sectional area.
5. Use the equivalent diameter (D)of the silo to determine the height of the silo required.

For shallow silos, $(h) \leq D \tan(45^0+\theta/2)$

$V_T = W_T/\rho$

$$V_T = \frac{\pi D^2 h}{4} = W_T/\rho$$

Replacing h in equation above,

$$W_T/\rho = \frac{\pi D^3 \tan\left(45+\frac{\theta}{2}\right)}{4}$$

$$D^3 = \frac{4W_T}{\rho\pi \tan\left(45+\theta/2\right)}$$

EXAMPLE

- A farmer wants to select a silo which would be used to store 100 tons of maize and 80 tons of soya bean. Consider the silo as a shallow silo.

Grain	Density kg/m³	Angle of repose
Maize	720	27
Soyabean	770	24

Solution

W_T = 100 tonnes = 100 x 10^3 kg

$V_T = 138.89 m^3$

$(h) \leq D \tan(45^0+\theta/2)$,

h is the height of grain in the silo

$h \leq 1.63D$

$$V_T = \frac{\pi D^2 h}{4}$$

$$V_T = \frac{\pi D^2 (1.63D)}{4}$$

D = 4.8m

$h \leq 1.63D$

$h \leq 7.8m$

5. CONVENTIONAL BAG STORAGE



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5. CONVENTIONAL BAG STORAGE

- The most common method of grain storage in many countries is bag storage.
- It may be done on raised platforms, in ventilated cribs, protected movable covers or in stores or warehouses.
-
- Warehouses, defined here, are facilities that provide a proper environment for the purpose of storing grains that require protection from elements
- Fig.3 gives the required warehouse management principles.

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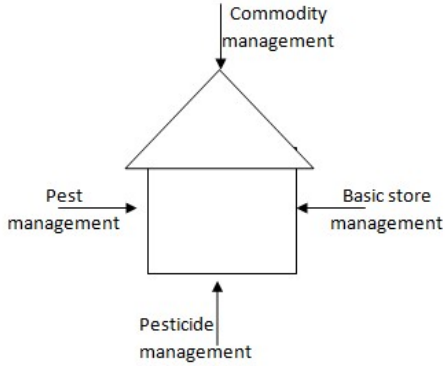


Fig.3 Factors to consider in warehouse management

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Different types of warehouses include:

- **Heated and unheated general warehouses**—provide space for bulk, rack, and bin storage, aisle space, receiving and shipping space, packing and crating space, and office and toilet space;
- **Refrigerated warehouses**—preserve the quality of perishable goods and general supply materials that require refrigeration. Includes freeze and chill space, processing facilities, and mechanical areas; and
- **Controlled humidity (CH) warehouses**—similar to general warehouses except that they are constructed with vapor barriers and contain humidity control equipment to maintain humidity at desired levels.

A. TYPES OF SPACES

- Depending on the programme of the warehouse being designed, space types may vary dramatically.
- Storage Space
- Office Space
- Loading Docks for shipping and receiving
- Light Industrial Space
- Computer Centers

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B. SPACE CONFIGURATIONS

- Warehouses should:
- Be designed based on current and future needs.
  - Facilitate changes in business/agency growth, and size/population of office and warehouse spaces within the building. Warehouse space should be easily adapted to new functions such as office (on ground or upper levels), computer centers, or light industrial/fabrication.
  - Accommodate need for future loading docks, truck space, and car parking spaces if space configuration changes through effective site design.

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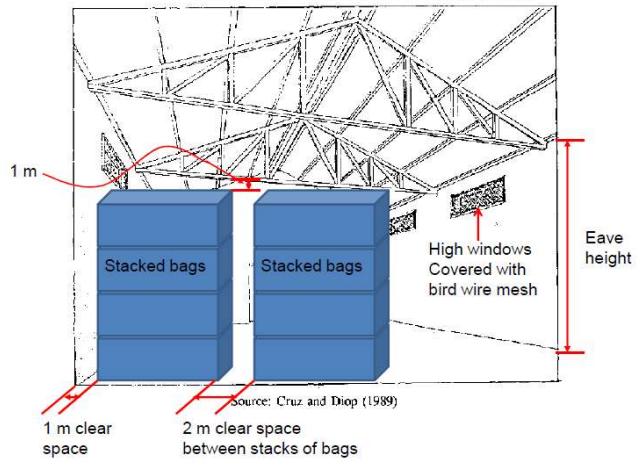
- Address material handling technologies and business practice, such as "just-in-time" storage, which have fundamentally changed operation of warehouses and distribution centers, and will continue to do so.
- Include roof design with built-in extra structural capacity to handle addition of future rooftop equipment.
- Be designed with fire protection capacity to accommodate storage of materials with a greater fire hazard, especially needed with high plastic product content or packaging, and plastic shrink-wrapped pallets.

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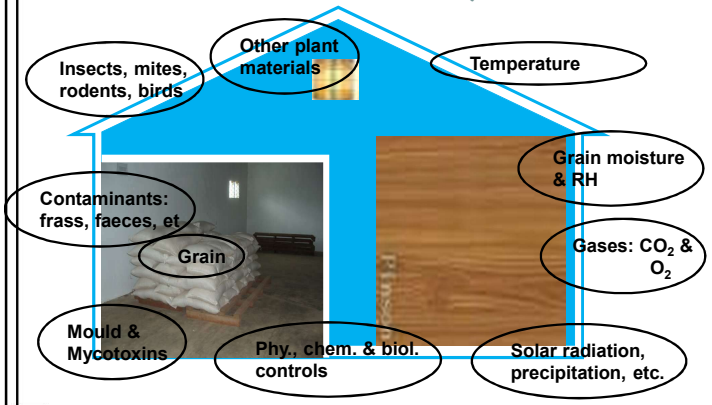
- Maximize utilization of space while providing adequate circulation paths for personnel and material handling equipment such as forklift trucks.
  - Use higher bays to take advantage of height allowances in the space.
- Optimize layout and configuration for the warehouse operation, including efficient circulation and material handling and storage processes.
- Relate interior and exterior receiving and shipping operations to the process flow of goods through the warehouse. Maximize utilization of space while providing adequate circulation paths for personnel and material handling equipment such as forklift trucks.

Bag Storage





The Stored Grain Ecosystem



5. CONVENTIONAL BAG STORAGE

- The Principal criteria required for proper storage in warehouses include:
  - Satisfactory control of humidity and temperature
  - Possibility of fumigation
  - Effective protection against rodents

5.1 Factors considered in the design of warehouses for structures

- Siting and orientation
- Warehouse should be constructed on a raised site
- Good drainage
- Firm soil for construction
- Good road connection to the warehouse
- Wind direction should be considered in the siting of the wall direction

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5.2 General construction features

Floors

- Floors should be raised at least 1m above the ground in other to prevent ground water from entering the store.
- They should serve as a vapour or moisture barrier.
- Concrete floors should be hard enough to bear the expected loads – this is to prevent cracks.
- Smooth surface floors help in cleaning.

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Walls

- Walls should have a smooth surface and should be without cracks
- Metal walls are usually not encouraged to be used.

Roofing

- The roof overhang should be at least 1m to ensure that the silo is kept in shade and its walls are protected against rain.

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Roofing

- There should also be properly sealed connection between the roofs and the walls as this will prevent birds from entering.
- In some cases, roof gutters are necessary for water collection and also to protect the foundation.

Doors

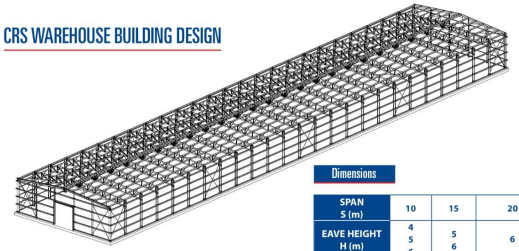
- Metal doors are usually used. If wooden doors are used, the lower part should be of metal.

Stacking of bags



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CRS WAREHOUSE BUILDING DESIGN



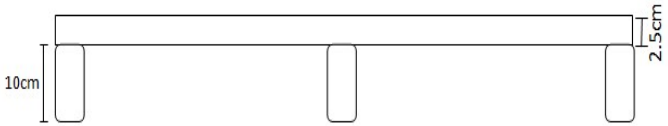
Dimensions

SPAN S (m)	10	15	20
EAVE HEIGHT H (m)	4 5 6	5 6	6
LENGTH L (m)	Structures with Bay spacing of 5 Meters can have indefinite length.		

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Stacking of bags

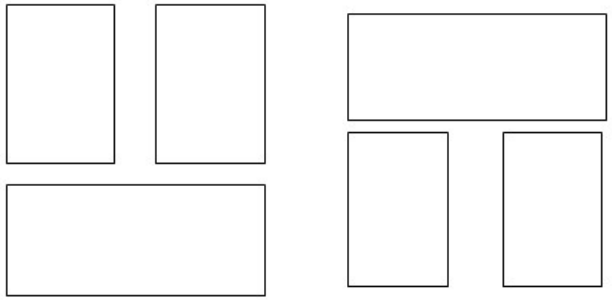
- It is done on pallets in range from 3, 5 and 8. Sacks being used for storage should be stacked in a way that allows air circulation.
- The pallets are usually kept 10cm off the ground.



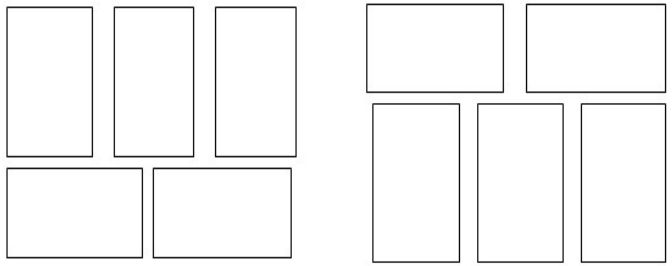
Packing of the bags is done from one end to the other end.

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3 Pack Arrangements



5 Pack Arrangements



Alternating bags placed lengthwise / widthwise

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- Space is provided between the wall and the stack and within the stacks
- The bags are placed on pallets
- The layers of bags are arranged perpendicularly and in a regular fashion
- The store is cleaned
- A space of at least 1m must be left between the top of a bag-stack and the roof for operations

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STORE CAPACITY

- In calculating the capacity of a store, the volume to be occupied by a given tonnage of grain is determined.
- The following information is required in the determination of the storage volume:
  - Useful surface area
  - Type of grain
  - Stowage factor
  - Bulk density (kg/m^3) or specific volume (m^3/kg)

Stowage factor is the reciprocal of bulk density.
Units: m^3 /tonne

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Food stuff	Stowage factor (m^3 /tonne)
Beans (bulk)	1.3
Millet (bags)	1.2
Maize (bulk)	1.4
Maize (bags)	1.6
Maize (heads or cobs)	2.2
Groundnut (seed in bags)	
Groundnut (in shells, bulk)	
Rice (paddy in bags)	

Steps for store capacity determination

- Determine the surface area
- Determine the useful surface area: Surface area- space occupied by gangways
- Determine useful volume by multiplying the useful surface area by the height of wall or expected height of product
- Determine the tonnage

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Example

- Consider a rectangular store 60m long by 12m wide and 3m high. There are two 1m wide gangways running lengthwise along the walls and a central gangway 4m wide running lengthwise. Also running widthwise are two 1m wide gangways along the walls plus another three 4m wide gangways. (Stowage factor 1.6, bag is 100kg)
- Assume bagged maize are stacked to the top of the walls determine the maximum number of bags of maize to be stored in this warehouse. Assume height of stack to be 3m

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Solution

Useful length: 60 -1-1- 4 – 4 - 4 = 46m

Useful width: 12- 1- 1- 4 = 6m

Useful surface area = 46m x 6m =276m^2

Useful volume = 276m^2 x 3m = 828m^3

Tonnage = Useful volume/storage factor

= 828m^3 / 1.6m^3/tonne = 517.5 tonnes

- Number of bags = 517.5 x 1000kg/ 100kg =5175bags

Note: The extent to which the available volume is used increase with the size of building.

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Utilisation Factor (UF)

- It is the proportion of floor usage in a warehouse. It is the ratio of the floor area covered with grain to the total floor area of the store.

UF (%) = FA_g/FA_t

FA_g =276m^2

FA_t =720m^2

UF = 38.33%

MANAGEMENT OF WAREHOUSES

1. Maintenance of equipment and warehouses
2. Correct handling and storage of the products
3. Management of the stored products
4. Regular control of the storage facilities
5. Keep correct records
6. Write reports regularly
7. Control of insects and rodents – treat building, repair cracks in walls, clean building
8. Bag stacking should be properly done to maximise space and maintain hygiene
9. Keep surrounding of warehouse clean

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6. VENTILATION/AERATION IN STORAGE

- Moisture movement is linked to the deterioration of products in storage (moulding).
- When grains are kept in storage heat and moisture are generated. These are undesirable as it will enhance the rate of deterioration of the product.
- Relative humidity of the air has to be checked in order not to exceed equilibrium relative humidity of the given moisture content. Air is therefore blown through stored grain (referred to as **Aeration**) to reduce moisture within he store.

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6. VENTILATION/AERATION IN STORAGE

Aeration is done to:

- Remove generated water
- Remove generated heat
- Maintain uniform environmental conditions

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AERATION FOR BULK STORAGE



6.1 DESIGN OF AERATION FOR BULK STORAGE

- Aeration design consist of determining the volume of air to be delivered per tonne minute and then designing the fan inlets, motor and system controls to deliver the desired air.

- The following steps are usually considered in the design of aeration of bulk storage example silos.
  1. Consider the design moisture content
  2. Consider heat that is generated as a result of water and carbon dioxide production.

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6.1 DESIGN OF AERATION FOR BULK STORAGE

3. Select the design day
4. Hours of operation of the ventilation system
5. Calculate in kilograms, the air needed per day.
6. Determine the volume flow rate and pressure
7. Select a fan required. (or number of fans)
8. Design of air distribution system
9. Design of power system

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6.2 AERATION IN WARE HOUSES/ BAG STORAGE

- The system usually used is refrigeration type system or dehumidification system.

- Cooling Unit: fans, evaporation, condensation